1. The three levels of management comprise the management
   a. society
   b. system
   c. strata
   d. hierarchy
   e. tenet

2. The importance of the function of _____ is based upon the fact that an organisation's goals are accomplished through its human resources.
   a. staffing
   b. organising
   c. controlling
   d. planning
   e. leading

3. In management, a comprehensive, fundamental law, doctrine, or assumption is called a(an):
   a. truism
   b. tenet
   c. hypothesis
   d. principle
   e. none of the above

4. All the following are basic management responsibilities *except*:
   a. organising the factors of production
   b. making business decisions
   c. taking risks
   d. eliminating risks
   e. none of the above

5. The number of subordinates a manager can effectively manage is referred to as:
   a. span of power
   b. span of subordination
   c. span of responsibility
   d. span of departmentation
   e. none of the above

6.Which of the following interpersonal roles played by the manager does not involve significant decision making?
   a. monitor
   b. leader
   c. spokesperson
   d. figurehead
   e. none of the above

7. Which of the following managerial levels suffers most from "identity problems"?
   a. supervisory management
   b. top management

c. middle management
d. hierarchical management
e. none of the above

8. During which of the following stages in the planning process is the company's mission statement developed?
   a. policy development
   b. definition of planning premises
   c. development of controls
   d. formulation of company objectives and strategic plans
   e. none of the above

9. The primary advantage of division of labour is based upon:
   a. specialisation
   b. concentration of authority
   c. concentration of responsibility
   d. conservation of resources
   e. none of the above

10. The process of allocating tasks to subordinates, providing authority to discharge those tasks, and exacting responsibility is called:
    a. delegation
    b. deputisation
    c. management by exception
    d. authorisation
    e. none of the above

11. Which of the following management function is the process of creating an organisation's structure by determining what tasks are to be done, who is to do them, how the tasks are to be grouped, who reports to whom, and where decision's are to be made?
    a. planning
    b. organising
    c. controlling
    d. leading
    e. staffing

12. Which of the following management functions includes motivating subordinates, directing others, selecting the most effective communication channels, and resolving conflicts
    a. staffing
    b. leading
    c. planning
    d. organising
    e. controlling

13. The first step in the organising process is:
    a. creating jobs
    b. grouping jobs into departments

c. creating authority/reporting relationships
d. determination of those tasks required to achieve objectives
e. none of the above

14. Which of the following is the *simplest* type of standing plan?
    a. an objective
    b. a policy
    c. a rule
    d. a procedure
    e. none of the above

15. People who have more freedom in their job performance are more satisfied with:
    a. participative leaders
    b. autocratic leaders
    c. benevolent autocrats
    d. exploitative autocrats
    e. none of the above

16. Robert Boateng has been asked to develop plans to ensure the efficient day-to-day use of resources allocated to her area of responsibility. Robert is creating:
    a. strategic plans
    b. tactical plans
    c. situational plans
    d. administrative plans
    e. none of the above

17. The primary purpose of control is to:
    a. reduce the size of the work force
    b. make decisions for management
    c. indicate when corrective action is needed
    d. make automation possible
    e. none of the above

18. All the following are barriers to delegation with the exception of
    a. Setting goals that are aligned
    b. Managers set unattainable or impossible goals.
    c. Managers overemphasise quantitative or easily measured goals.
    d. Managers mistakenly reward ineffective goal setting.
    e. Managers fail to reward effective goal setting.

19. What is the result of maximum delegation?
    a. Decentralisation
    b. Democratisation

c. Centralisation
d. Industrialisation
e. Departmentalisation

20. In _____ decision-making, control of rewards, punishment and authority for determining policy procedures for achieving goals lies with the leader.
    a. free style leadership style
    b. authoritarian leadership style
    c. democratic leadership style
    d. participative leadership style
    e. none of the above

21. Abraham Maslow's Hierarchy of Needs includes the following, except
    a. psychological needs
    b. self-actualisation needs
    c. belongingness needs
    d. technological needs
    e. safety needs

22. All the following quantitative forecasting techniques make use of historical data, with the exception of
    a. simple moving averages
    b. weighted moving averages
    c. causal modelling
    d. naive Approach
    e. exponential smoothing

23. Preliminary control focuses on _____
    a. controlling ongoing processes
    b. preventing deviations
    c. using outputs to control inputs to future processes
    d. both inputs and outputs
    e. none of the above

24. The first use of standard costing was to control……
    a. manufacturing costs
    b. administrative costs
    c. supplier management cost
    d. human resource cost
    e. none of the above

25. At the breakeven point in breakeven analysis, _____
    a. total revenue is equal to total cost
    b. total revenue is equal to the total variable cost
    c. total revenue is equal to the total fixed cost
    d. the total cost is equal to the total variable cost
    e. total revenue is equal to the price per unit

26. Which of the following is a variable cost?
    a. rents
    b. insurance premiums
    c. raw material cost
    d. leased equipment
    e. marketing and Advertising costs

27. In order to estimate the breakeven point, the Operations Manager needs to estimate the following, except:
    a. variable costs
    b. fixed cost
    c. demand for the goods
    d. the time it takes to produce one unit
    e. price for the goods

28. A decision-maker under conditions of risk knows
    a. With certainty the outcome of any decision or alternative
    b. The probability of occurrence of the outcomes or consequences for each choice
    c. The optimistic outcome and the pessimistic outcomes of any decision or alternative
    d. The maximin, maximax and Laplace outcomes of any decision or alternative
    e. Nothing about the outcome of a decision or alternative

29. When using the Maximax decision criterion, the decision maker
    a. evaluates the minimum outcomes and selects the maximum
    b. evaluates the maximum outcomes and selects the minimum
    c. evaluates the maximum outcomes and selects the maximum
    d. finds the expected monetary value of the decision or alternative
    e. finds the average of all possible outcomes

30. All of the following are advantages offered by robots over human workers *except*:
    a. lower costs
    b. higher quality
    c. greater work reliability
    d. ability to bring creativity to the job
    e. none of the above

31. In the management hierarchy of a university, a dean is best described as a(n):
    a. academic manager
    b. top manager
    c. supervisor
    d. middle manager
    e. none of the above

32. Emmanuel Fynn is preparing to make one of the most important decisions of her career. He will begin by:
    a. evaluating her assets
    b. evaluating the decision situation
    c. analyzing alternatives.
    d. determining results.
    e. formulating goals

33. In general, _____ tend to be characterised by greater communication between upper and lower-level management, an increased capacity to respond to the environment, and lower total managerial costs.
    a. flat organisations
    b. tall organisations
    c. narrow organisations
    d. wide organisations
    e. none of the above

34. The process of locating, identifying, and attracting capable applicants is known as
    a. personnel planning
    b. selection
    c. appraisal
    d. recruitment
    e. none of the above

35. Which of the following types of decisions are used to solve non-recurring problems?
    a. partially structured decisions
    b. non programmed decisions
    c. programmed decisions
    d. procedural decisions
    e. none of the above

36. Repeated resolutions with the same data will yield the same results in a(an):
    a. partially structured decision
    b. ill structured decision
    c. repetitive decision
    d. programmed decision
    e. none of the above

37. Mary Mansah meets with her marketing and sales executives each month to develop new product ideas, plan new product introductions, and review the firm's advertising strategy. This committee is best described as a(n):
    a. ad hoc committee
    b. formal committee
    c. informal committee
    d. temporary committee
    e. none of the above

38. Jim Johnson has been asked to decide whether her firm should market a new artificial sweetner. Jim knows that many similar products have been withdrawn from the market in recent years as a result of governmental action. Jim's decision is best described as a decision under
    a. certainty
    b. risk
    c. uncertainty
    d. control
    e. none of the above

39. Which of the following serves as the foundation for the planning process?
    a. the grand strategy
    b. tactical planning
    c. the organisational mission
    d. the environmental statement
    e. none of the above

40. Human resources planning involves all the following except:
    a. evaluating existing labour sources
    b. forecasting future labour needs
    c. ensuring labour union representation
    d. ensuring the availability of workers when needed
    e. none of the above

41. The attempt to organise work to reduce effort, increase employee satisfaction, reduce costs, and boost efficiency is called:
    a. job design
    b. job analysis
    c. job description
    d. job specification
    e. none of the above

42. The first activity to undertake at the beginning of the management process is
        a. coordinating resources
        b. planning
        c. goal setting
        d. monitoring
        e. conduct a SWOT analysis

43. The fifth concept in organising is
        a. designing jobs,
        b. managing line and staff positions
        c. establishing the span of management
        d. grouping jobs,
        e. dealing with authority and responsibility

44. The idea that high levels of specialization and standardization would increase efficiency was proposed by
    a. Adam Smith
    b. Abraham Lincoln
    c. Karl Marx
    d. Frederick Taylor
    e. Friedrich Hayek

45. The leading function of management includes all the following *except*:
    a. motivating employees towards goal attainment
    b. grouping activities and resources in a logical and appropriate manner
    c. managing group processes
    d. managing conflicts
    e. managing change

46. The third step in the delegation process is
    a. grant authority
    b. create accountability
    c. demand transparency
    d. allocate resources
    e. assign responsibility

47. A manager is said to be effective if :
    a. he/she is able to use minimum resources to achieve organisational goals
    b. he/she is able to convince his subordinates to agree with his ideas
    c. he/she is able to achieve the goals set for the organisation
    d. he/she follows his instincts
    e. none of the above

48. The two key concepts of organising include organisational
    a. design and structure
    b. division of labour and departmentation
    c. job rotation and job enlargement
    d. production and finance
    e. none of the above

49. The process of deciding what objectives to pursue during a future time period and what to do to achieve those objectives is
    a. controlling
    b. planning
    c. leading
    d. staffing
    e. none of the above

50. Which of the following refers to broad, general guides to action that constrain or direct objective attainment?
    a. rules
    b. policies
    c. procedures
    d. managerial guidance
    e. none of the above

51. The arrangement of human and physical resources in a structure designed to co-ordinate and channel effort toward the accomplishment of the firm's strategic plan is referred to as
    a. management
    b. planning
    c. organising
    d. leading
    e. none of the above

52. In management, the two sources of power include
    a. organisational power and personal power
    b. legitimate power and expert power
    c. reward power and referent power
    d. leading power and expert power
    e. none of the above

53. Which of the following represents a deliberate arrangement of people to accomplish some specific purpose?
    a. hierarchy
    b. organising
    c. an organisation
    d. departmentation
    e. none of the above

54. Which of the following interpersonal function of management is concerned with interacting with peers and other people outside the organisational unit to gain information and favours?
    a. figurehead
    b. leader
    c. liaison
    d. negotiator
    e. none of the above

55. Which of the following informational function of management is concerned with transmitting special information within the organisation?
    a. monitor
    b. disseminator
    c. spokesperson
    d. liaison
    e. none of the above

56. The “heart and soul” of management is
    a. planning
    b. organising
    c. leading
    d. controlling
    e. none of the above

57. Two managerial activities that cannot be separated are _____ and _____.
    a. planning and organising
    b. planning and decision making
    c. organising and directing
    d. leading and directing
    e. controlling and leading

58. In which year did Graicunas, a French management consultant attempt to demonstrate mathematically the increase in the number of relationships arising from each increase in the number of subordinates reporting to the manager.
    a. 1776
    b. 1976
    c. 1944
    d. 1933
    e. none of the above

59. According to the formula developed by Graicunas, a superior who has 14 subordinates will face _____ possible superior-subordinate relationships.
    a. 14,496,582
    b. 229,740
    c. 114,870
    d. 196
    e. 14

60. The institution of a quality assurance system is most consistent with which of the following basic management functions?
a. planning
b. organising
c. controlling
d. leading
e. none of the above

61. Which of the following is money due from an individual or another business as payment for the performance of services or the sale of goods on credit?
    a. account receivable
    b. account payable
    c. bond
    d. note
    e. none of the above

62. A long-term solvency ratio which indicates the company's ability to meet its long term financial obligations is known as
    a. debt to equity ratio
    b. return on equity ratio
    c. acid Test ratio
    d. equity ratio
    e. none of the above

63. In a typical finance and accounting function, who would have responsibility for inventory valuation, budgetary control, variance analysis and project appraisal?
    a. the Management Accountant
    b. the Financial Accountant
    c. the Internal Auditor
    d. the Treasurer
    e. none of the above

64. A financial ratio that measures the organisation's internal performance with respect to key operations defined by management is a/an:
    a. activity ratio
    b. leverage ratio
    c. liquidity ratio
    d. performance ratio
    e. none of the above

65. With the payment of a liability by cheque:
    a. the asset of bank is increased and the liability of creditor is reduced
    b. the asset of bank is reduced, and the liability of the debtor is reduced
    c. the asset of bank is reduced, and the liability of the creditor is reduced
    d. the asset of bank is increased, and the liability of the debtor is increased
    e. none of the above

66. If profit is to be maximised, management must minimise the proportion of capital invested in:
    a. current assets
    b. fixed assets
    c. stock market
    d. retained earnings
    e. none of the above

67. The difference between total current assets and total current liabilities is known as
    a. capital
    b. accounts payable
    c. asset-based capital
    d. working capital
    e. none of the above

68. Which of the following refers to probable obligations of the entity that result from past transactions, which will be paid with assets or services?
    a. capital
    b. franchise
    c. liabilities
    d. face amount
    e. none of the above

69. Which the following ratios measure the magnitude of owners and creditors' claims on the organisation and indicate the organisation's ability to meet long-term obligations?
    a. current ratio
    b. profitability ratio
    c. leverage ratio
    d. operating ratio
    e. none of the above

70. Which of the following represents a standardised format that organisations use to accumulate the monetary effects of transactions on each financial statement?
    a. Account
    b. Accounting
    c. Book Keeping
    d. Notes
    e. none of the above

71. When the owner of a business takes money out of the bank for his own use then
    a. there is a decrease in asset (Bank) and a decrease in capital
    b. there is a decrease in asset (Bank) and an increase in capital
    c. there is a decrease in asset (Bank) and a decrease in liability
    d. there is a decrease in asset (Bank) and an increase in liability
    e. none of the above

72. Which of the following represents revenues that have been earned and expenses that have been incurred by the end of the current accounting period but will not be collected or paid until a future accounting period?
    a. accumulated Other Comprehensive
    b. accruals
    c. decelerated Bond
    d. doubtful Accounts
    e. none of the above

73. The main aim of accounting is
    a. to produce a trial balance
    b. to maintain ledger accounts for every asset and liability
    c. to provide financial information to users of such information
    d. to record every financial transaction individually
    e. none of the above

74. A legal document given by a borrower to a lender stating the time of repayment and directly, or indirectly, the amount to be repaid is referred to as
    a. Note
    b. Bond
    c. Debenture
    d. Inbenture
    e. none of the above

75. Which of the following represents shares in a corporation's equity that usually do not give stockholders voting rights but grant specific dividends subject to various guarantees and preferences.?
    a. outstanding stock
    b. preferred stock
    c. common stock
    d. debenture
    e. none of the above

76. Accounting and reporting to satisfy the inside demand for accounting information is called:
    a. financial accounting
    b. management accounting
    c. book keeping
    d. inside satisfaction
    e. none of the above

77. In business, ***Other People's Money*** is commonly called
    a. operating money
    b. activity finance
    c. monetary booster
    d. financial leverage
    e. none of the above

78. The component of equity in a corporation representing accumulated profits in excess of losses and payments to owners is referred to as
    a. retained earnings
    b. cash flow
    c. capital
    d. profit before tax
    e. none of the above

79. An analytical tool designed to identify significant relationships; measures the proportional relationship between two financial statement amounts is a(an)
    a. ratio analysis
    b. analytical security
    c. business percentage checks
    d. financial book keeping
    e. none of the above

80. Manufacturing equipment and land are classified as:
    a. inventory items
    b. short-term assets
    c. capital outlay items
    d. liquid assets
    e. none of the above

81. Which of the following financial statements is used to analyze a firm's assets, liabilities, and net worth position of a given point in time?
    a. the liquidity report
    b. the income statement
    c. the balance sheet
    d. the ratio statement
    e. none of the above

82. A systematic and rational allocation of the acquisition cost of an intangible asset over its useful life is known as
    a. accrual
    b. amortisation
    c. debenture
    d. depreciation
    e. none of the above

83. Traditional accounting data are weakest on which of the following criteria for standards
    a. validity of the information
    b. timeliness of the information
    c. reliability of the information
    d. appropriateness of the units of measurement
    e. none of the above

Carefully study the following Income Statement and Statement of Balance Sheet for Obaa Yaa & Daughters Co. Ltd. for the year 2013 and answer questions 84 through 91. You are required to show how you obtain your answer, please.

Obaa Yaa & Daughters Co. Ltd.
Income Statement
For the Year Ended 31 December 2013

REVENUE	$	$
Sales	325,200	
Interest	600	
Rent	1,200	
Total Revenues		327,000
EXPENSES		
Cost of goods sold	117,000	
Selling	113,400	
General and administrative	34,600	
Interest	3,000	
Total expenses before income tax		268,000
Income before extraordinary item and tax		59,000
Income taxes on income before extraordinary items ($59,000 x 0.40)		23,600
Income before extraordinary items		35,400
Extraordinary loss	9,000	
Less tax savings ($9000 x 0.40)	3,600	5,400
NET INCOME		30,000

Obaa Yaa & Daughters Co. Ltd.
Balance Sheet as at 31 December 2013

ASSETS	$	$
Fixed Assets		
Land		8,000
Building	160,000	
Accumulated depreciation, building	(100,000)	60,000
Equipment	91,000	
Accumulated depreciation, equipment	(36,000)	55,000
Total fixed assets		123,000
Current Assets		67,300
Cash		
Accounts receivable	45,000	
Allowance for doubtful accounts	2,200	42,800
Notes receivable		8,000
Interest receivable		100
Inventory		90,000
Prepaid expenses		400
Total current assets		208,600
Total assets		331,600
STOCKHOLDERS' EQUITY		
Contributed capital:		
Common stock: par 10%, 15,000 shares issued and outstanding	150,000	
Contributed capital in excess of par	20,000	
Retained earnings	61,500	
Total stockholders' equity		231,500
LIABILITIES		
Current liabilities		
Accounts payable		29,000
Interest payable		500
Rent collected in advance		600
Income tax payable		20,000
Total current liabilities		50,100
Long term liabilities		
Bonds payable, 6 %		50,000
Total liabilities		100,100
TOTAL STOCKHOLDERS' EQUITY AND LIABILITIES		331,600

84. Calculate the current ratio.
    a. 6.56
    b. 4.16
    c. 3.49
    d. 3.12
    e. none of the above

85. Compute the quick ratio.
    a. 2.34
    b. 2.36
    c. 2.37
    d. 2.38
    e. none of the above

86. Compute the equity multiplier.
    a. 1.29
    b. 1.54
    c. 1.43 total assets/total owners' equity
    d. 1.75
    e. None of the above

87. Determine the debt ratio.
    a. 0.20
    b. 0.30 total liability/total stockholders' equity and liabilities
    b. 0.40
    c. 0.15
    d. none of the above

88. Compute the debt to equity ratio.
    a. 0.52
    b. 0.32
    c. 0.43 total liability/owners' equity
    d. 0.25
    e. none of the above

89. Compute the return on equity.
    a. 0.13 net income/total owners' equity
    b. 0.15
    c. 0.17
    d. 0.19
    e. none of the above

90. Compute the equity ratio.
    a. 0.40
    b. 0.50
    c. 0.60
    d. 0.70
    e. none of the above

91. Calculate the earnings per share in $.
    a. 2.13
    b. 1.19
    c. 2.00
    d. 1.54
    e. none of the above

92. Given the following details, compute the Inventory turnover ratio.

Net Sales	= $30,000
Cost of Goods Sold	= $117,000
Total Current Assets	= $208,600
Average Inventory	=$90,000

    a. 3.00
    b. 1.30
    c. 1.78
    d. 3.2
    e. none of the above

93. Project management is the _____of events that, together, comprise the project.
    a. planning and decision making
    b. planning and organising
    c. planning and control
    d. management
    e. none of the above

94. Two forms of project networks are used. They are:
    a. AOA and AON
    b. AOA and NOA
    c. CPM and PERT
    d. NODES and ARCS
    e. Gantt charts and network diagrams

95. Project managers use PERT and CPM network procedures to help them answer all these questions, except:
    a. What is the total time to complete the project?
    b. What are the scheduled start and finish dates for each specific activity?
    c. Will the project be completed to the specifications of the clients?
    d. How long can a “non-critical” activity be delayed before they can cause a delay in the total project?
    e. Which activities are critical and must be completed exactly as scheduled in order to keep the project on schedule?

96. Why are Dummy activities used in drawing network diagrams
    a. They are used to show precedence relationships in AON network diagrams
    b. They are used to show precedence relationships in AOA network diagrams
    c. They are used to show activities that have 0 duration in network diagrams
    d. They are used to connect the dots network diagrams used in project management
    e. They are used to show how the project proceeds from beginning to end

97. PERT network method is used to schedule projects that
    a. Have thousands of tasks that have task times that are certain.
    b. Have a small number of tasks that are sequential in order
    c. Have a large number of tasks that have uncertain task times
    d. Have small of tasks that have task times that are certain.
    e. Have thousands of tasks that have expected task times known.

98. The following are examples of projects except:
    a. research and development of new products and processes
    b. construction of a new Auditorium
    c. shutdown and maintenance of a plant
    d. production of a model of a car
    e. design and installation of a new layout

99. Project planning starts with the development of a **Statement of Work (SOW)** which is:
    a. a definition of the individual tasks that must be performed
    b. a description of the objectives to be achieved and the work to be done.
    c. a list of resources needed to complete the project
    d. an estimate of the time required to perform each activity
    e. a detailed description of the work to be done by the project initiators

100. PERT/CPM concentrates on the time aspect of a project and provides information that can be used to
    a. schedule and control individual activities so that the entire project is completed on time
    b. budget for the project so that if actual costs are exceeding budgeted costs, corrective action may be taken to keep the costs within the budget.
    c. determine the members of the project team
    d. select the suppliers to supply the materials and equipment for the project
    e. determine the type of project organization to use for the project

101. A network diagram not only depicts the activities that must be completed to bring the project to an end but it also portrays:
    a. the duration of the activities
    b. the start and stop times of the activities
    c. the predecessor relationships among the activities
    d. event or milestones of the project
    e. all of the above

Use the information below on a project to answer the questions 102 – 111 that follow:

Activity	Optimistic time	Most likely time	Pessimistic Time	Immediate Predecessors
A	5	6	7	_
B	10	13	28	_
C	1	2	15	A
D	8	9	16	B
E	25	36	41	B, C
F	6	9	18	D

102. What project scheduling method will you use?

a. Gantt Charts
b. Time /Cost trade Offs
c. Critical Path Method
d. Program Evaluation and Review Technique
e. Network Diagrams

103. What are the expected time and variance for activity A?

a. 6 and 0.11
b. 35 and 7.11
c. 15 and 9.00
d. 10 and 1.77
e. 15 and 1.78

104. What are the expected time and variance for activity B?

a. 6 and 0.11
b. 15 and 1.78
c. 15 and 9.00
d. 35 and 7.11
e. 10 and 1.77

105. What are the expected time and variance for activity C?

a. 6 and 0.11
b. 15 and 9.00
c. 35 and 7.11
d. 10 and 1.77
e. 4 and 5.44

106. What are the expected time and variance for activity D?
   a. 6 and 0.11
   b. 10 and 5.44
   c. 10 and 1.77
   d. 15 and 1.78
   e. 35 and 7.11

107. What are the expected time and variance for activity E?
   a. 6 and 0.11
   b. 15 and 1.78
   c. 10 and 1.77
   d. 4 and 5.44
   e. 35 and 7.11

108. What are the expected time and variance for activity F?
   a. 6 and 0.11
   b. 10 and 4.00
   c. 10 and 5.44
   d. 35 and 7.11
   e. none of the above

109. What is the project's critical path?
   a. B, D and E
   b. B and E
   c. A, C and E
   d. B, D and F
   e. Cannot be determined

110. What is the sum of the variances on the critical path?
   a. 13.00
   b. 17.89
   c. 20.00
   d. 16.11
   e. 27.11

111. What is the expected project duration?
   a. 45 days
   b. 35 days
   c. 50 days
   d. 75 days
   e. 55 days

112. A quick, rough estimate of the time required for money to double can be obtained by dividing which of the following numbers by the compound interest rate?
    a. 100
    b. 72
    c. 64
    d. 52
    e. none of the above

113. Items which are not easily expressed in terms of dollars are called:
    a. indirect costs
    b. variable costs
    c. intangible costs
    d. legal issues
    e. none of the above

114. Interest that is calculated using only the principal is called:
    a. simple interest
    b. effective interest
    c. add-on interest
    d. compound Interest
    e. none of the above

115. If $1000 is borrowed at 10% per year simple interest, the total amount due at the end of five years is nearest to:
    a. $1,100
    b. $1,250
    c. $1500
    d. $1611
    e. none of the above

116. The amount of money five years ago that is equivalent to $1000 now at 10% per year compound interest is nearest to:
    a. $621
    b. $667
    c. $1500
    d. $1611
    e. none of the above

117. The symbol that represents a uniform cash flow in consecutive interest periods is:
    a. F
    b. P
    c. G
    d. A
    e. none of the above

118. Minimum attractive rate of return refers to:
   a. the rate of return achievable from a safe government bond.
   b. the rate of return that must be equalled or exceeded to render an investment attractive.
   c. the rate of return guaranteed on a bank savings account.
   d. the lowest rate of return required to attract foreign capital.
   e. none of the above

119. All of the following are examples of cash inflows except:
   a. income taxes
   b. asset salvage value
   c. operating cost reduction
   d. construction cost savings
   e. none of the above

120. All of the following are interest rates except:
   a. Return on investment
   b. MARR
   c. Accrued interest
   d. Rate of return
   e. None of the above

121. When the interest rate is 10% per year, all of the following are equivalent to $5,000 now except:
   a. $4,545 one year ago.
   b. $5,500 one year hence.
   c. $4,021 two years ago.
   d. $6,050 two years hence.
   e. None of the above

122. In the uniform series factors, which involve the symbol A and P, the P value is always located _____.
   a. two and three
   b. sinking fund
   c. one interest period ahead of the first A value
   d. coincident with the A value
   e. none of the above

123. A uniform series of deposits intended to accumulate to a given future amount of money is known as a (an):_____
   a. sinking fund
   b. equivalent pool
   c. base amount
   d. first A value
   e. none of the above

124. A cash flow series which increases by a constant percentage is known as a (an):
   a. arithmetic series
   b. geometric series
   c. compound interest rate
   d. base amount
   e. none of the above

125. In all interest factors, the rate must be a _____.
   a. simple interest rate
   b. compound interest rate
   c. either simple interest or compound interest rate
   d. whole number
   e. none of the above

126. The equation that is used to calculate a present worth from a single future amount is:
   a. $P = F (1 + i)^n$
   b. $P = F [1 / (1 + i)^n]$
   c. $P = F [1 / (1 + i)^n]- 1) / i((1 + i)^n)]$
   d. $P = F [(1 + i)^n]- 1) / i]$
   e. none of the above

127. The operating cost of a small machine is $800 in year one, $900 in year two, $1000 in year three, increasing by $100 per year through year ten. At an interest rate of 8% per year, the equivalent annual worth of the machine is nearest to:
   a. $1,187
   b. $2,598
   c. $1,149
   d. $7,966
   e. none of the above

128. A young couple wishing to save money for their child's first year in college purchases an insurance policy that will yield $10,000 fifteen years from now. The cost of the policy is $500 per year for 15 years, beginning one year from now. The rate of return on their investment is nearest to:
   a. 3%
   b. 4%
   c. 5%
   d. 6%
   e. none of the above

129. The length of time required for money to quadruple in value at an interest rate of 6% per year is nearest to:
   a. 12 years
   b. 18 years
   c. 24 years
   d. 30 years
   e. none of the above

130. A deposit of $10,000 now at 10% per year interest will accumulate to how much twenty years from now?

a. $30,000
b. $46,275
c. $67,275
d. $83,195
e. none of the above

131. By doing annual updating of a certain production line, a manufacturing company can avoid spending $100,000 for a new system ten years from now. At an interest rate of 10% per year, the annual amount the company could afford to spend beginning one year from now is nearest to:

a. $6,275
b. $16,270
c. $24,190
d. $38,550
e. none of the above

132. At an interest rate of 10% per year, annual deposits of $1,000 in years one through ten would accumulate to how much immediately after the last deposit?

a. $22,819
b. $19,062
c. $15,937
d. $6,145
e. none of the above

133. If a company wanted to make a single investment now instead of spending $20,000 five years from now, how much would the investment be at an interest rate of 10% per year?

a. $32,300
b. $12,418
c. $5,276
d. $3,276
e. none of the above

134. If a company invests $25,000 in new packaging equipment, by how much must it reduce its annual costs if it expects to recover the investment in seven years at an interest rate of 10% per year?

a. $48,718
b. $12,830
c. $5,135
d. $2,635
e. none of the above

135. In order to update a production process, a company can spend money now or four years from now. If the amount now would be $20,000, what equivalent amount could the company spend four years from now at an interest rate of 10% per year?

a. $63,380
b. $47,690
c. $35,620
d. $29,282
e. none of the above

136. If a small company invests its annual profits of $150,000 in a stock fund which earns 18% per year, the amount in the fund after ten years will be nearest to:

a. $3,528,000
b. $2,153,000
c. $785,000
d. $479,000
e. none of the above

137. What is the equivalent amount in year ten of an expenditure of $5,000 in year one, $6,000 in year two, and amounts increasing by $1,000 per year through year ten? Assume the interest rate is 10% per year.

a. $139,060
b. $92,169
c. $53,614
d. $30,723
e. None of the above

138. A short-haul trucking company purchased a used dump truck for $12,000. The company paid $5,000 down and financed the balance at an interest rate of 10% per year for five years. The amount of its annual payment is nearest to:

a. $4,346
b. $3,166
c. $1,846
d. $1,447
e. none of the above

139. In order to have money for their son's college education, a young couple started a savings plan into which they made intermittent deposits. They started the account with a deposit of $2,000 (in year zero) and then added $3,000 in years two, five and six. The amount they had in the account in year ten if they earned interest at 12% per year was nearest to:

a. $23,647
b. $20,913
c. $17,320
d. $15,170
e. none of the above

140. In the derivation of the F/A factor, the F is located one period after the last A.
a. False
b. True

141. In the interest formulas, a simple interest can be used in any of the single payment formulas as long as the n values correspond to the interest period.

a. False
b. True

142. An arithmetic gradient refers to cash flow wherein the values change by the same percentage in each interest period.
a. False
b. True

143. In calculating the present worth of an arithmetic gradient series, the only difference between an increasing and a decreasing gradient calculation is the minus sign for the base amount.
a. False
b. True

144. In solving from an unknown interest rate involving only the F/P formula, it is possible to solve for *i* directly by re-arranging the equation.
a. False
b. True

145. A nominal interest is similar to a(n):
a. compound interest rate.
b. effective interest rate.
c. inflated interest rate.
d. simple interest rate.
e. none of the above

146. When no compounding period is given with an interest rate, the compounding period:
a. Cannot be determined from the statement.
b. Is assumed to be monthly.
c. Is equal to the interest period.
d. Must be less than interest period.
e. none of the above

147. When interest is compounded continuously:
a. The interest rate must be converted into an annual rate.
b. The cash flow must be converted into continuous cash flow.
c. The cash flow must also occur continuously.
d. none of the above
e. all the above

148. If you deposit $1000 per month into an account which pays interest at a rate of 12% per year compounded monthly, the amount of money you would have at the end of five years is nearest to:

a. $81,670
b. $76,234
c. $68,321
d. $6,353
e. none of the above

149. The owner of a small business borrowed $70,000 with an agreement to repay the loan with quarterly payments over a five year time period. If the interest rate is 12% per year compounded quarterly, his loan payment each quarter is nearest to:

a. $19,419
b. $9,372
c. $5,864
d. $2,605
e. none of the above

150. A metal plating company wants to set aside money now to prepare for a lawsuit it expects to face in four years. If the company wants to have $1,000,000 available at that time, how much must it set aside now in one lump sum if the account will earn 1% per month?

a. $961,000
b. $836,400
c. $711,800
d. $620,300
e. none of the above